Gesticulator

Motion Control M4L Device

User Manual

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1. Setup

1. Download the **ZIG SIM** app from the Apple App Store or Google Play store to the device you will be using for motion capture.
2. Find the local IP address of the computer that will be running Gesticulator.
   a. **Windows:**
      i. Open command prompt.
      ii. Type **ipconfig** and press enter.
      iii. The address next to **IPv4 Address** is your local IP address.
   b. **OSX:**
      i. In the Spotlight search, type **Terminal** and hit enter.
      ii. Type **ipconfig getifaddr en0** and hit enter.
      iii. Your local IP address will be displayed in Terminal.
3. For first time setup of ZIG SIM, go to the Settings tab and configure the settings as follows:

   Data Destination → **Other App**
   Protocol → **UDP**
   IP Address → **local IP address**
   Port Number → **50000**
   Message Format → **OSC**
   Message Rate (per sec) → **30**
   Device UUID → **1234**

   - After this initial setup, you should not need to enter these settings again.
4. Go to the **Sensor** tab in ZIG SIM and enable the following sensors:
   1. ACCEL
   2. GRAVITY
   3. GYRO
   4. QUATERNION
   5. COMPASS

6. 2D TOUCH
5. Place Gesticulator on any audio or MIDI track in Ableton Live.
6. In ZIG SIM, open the **Start** tab to begin sending your device's motion data to Gesticulator.
7. If configuration is successful, the settings in the top half of ZIG SIM should be colored green with data from each sensor flowing below it, the axis indicators in Gesticulator should be lit up and receiving data, and the visualizations should be colored and moving.

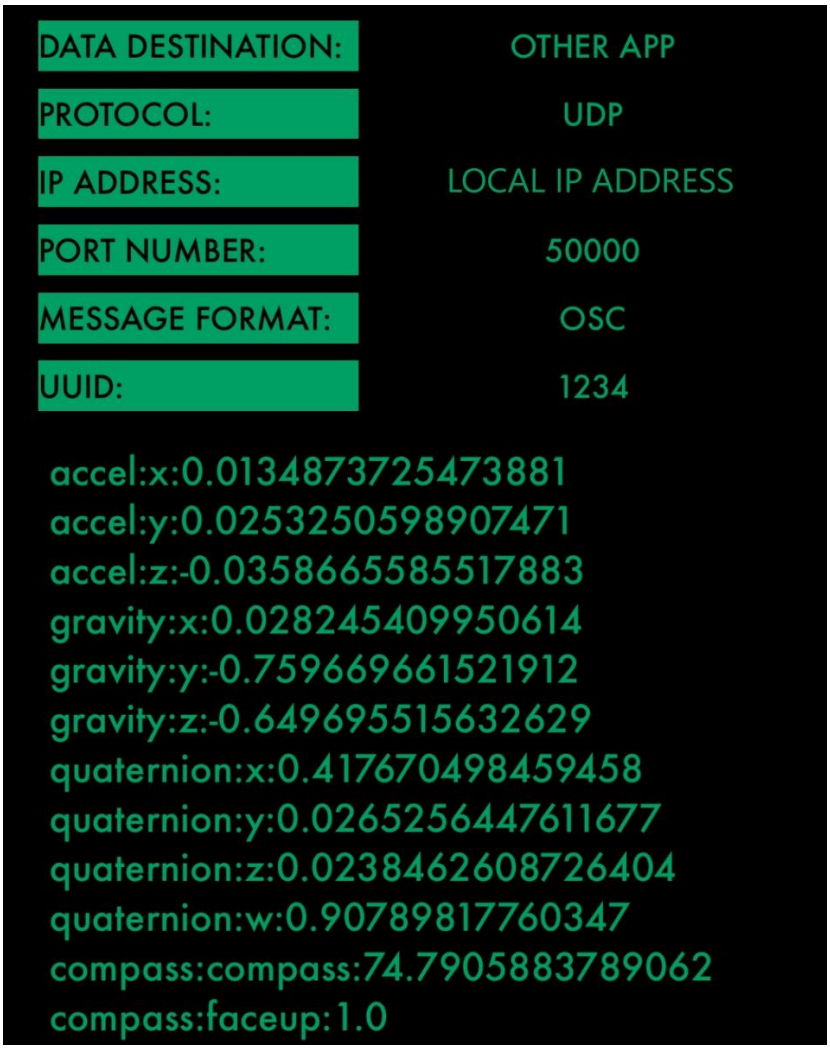

Fig. 1 ZIG SIM running successfully. Note that the touch sensor's input will not appear in the list of sensor inputs unless you are actively touching the screen.

2. How to Use

After the initial setup, running Gesticulator is simple:

1. Open ZIG SIM on your smartphone and open the **Start** tab to begin sending your device's motion sensor data to Gesticulator.
2. Place Gesticulator on any MIDI or audio track in Ableton Live.
3. Begin mapping!

Mapping:

Gesticulator has an individual map button assigned to each axis of every sensor (for more information on each sensor's functions, see page 7).

To map a sensor's axis to a parameter in Ableton:

1. Click the desired **Map** button so it begins flashing, indicating it is ready to map to a parameter.
2. Click the desired parameter to map.

The map button should now display the parameter's name, indicating that it is successfully mapped.

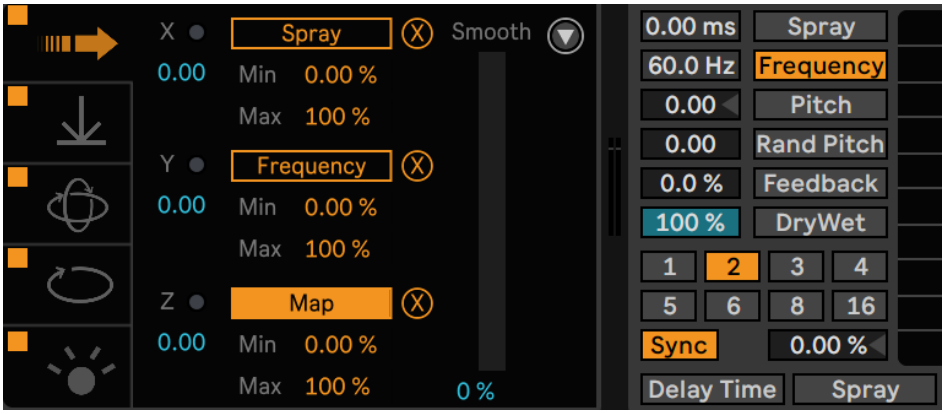

Fig. 2 The x and y-axes are mapped to Spray and Frequency, while the z-axis is ready to map.

To clear the mapped parameter, click the X button next to the parameter. The map button should now display **Map** again, showing it is ready for a new parameter.

Configuring:

Once a parameter is mapped, it can be configured in three ways:

1. The minimum value can be set with the **Min** parameter underneath its respective map button.
   - This is a percentage of the maximum value of the parameter.
2. The maximum value can be set with the respective **Max** parameter.
   - Configuring the Min and Max values allows for a range to be defined for each mapped parameter.
3. The **Smooth** slider is used to smooth out the incoming signal for each respective sensor. Use this to create slower, more gradual changes to the mapped parameter.

This diagram will explain the primary functions of Gesticulator.

Fig. 3 The primary UI of Gesticulator. Each label is named and described below.

Label	Name	Function
1	Input button	Turn on/off receiving input from all sensors.
2	Port button	Set the port number to match the sending device.
3	Help button	Quick configuration instructions for ZIG SIM.
4	Tab selection	Select a sensor to view. Turn on/off with the buttons.
5	Axis value	Displays the live value of each sensor's axes.
6	Axis indicator	Lights up when actively receiving sensor input.
7	Map	Click to map, clear the mapping, and set min/max values.
8	Smooth	Apply smoothing to the incoming signal for each sensor.
9	Show display	Show or hide the visualizer for each sensor.
10	Visualizer	Display the live reading of each sensor.

3. How Gesticulator Works

3.1 OSC

"OpenSoundControl (OSC) is a data transport specification (an encoding) for realtime message communication among applications and hardware" developed at CNMAT.[1] In other words, OSC is a way of formatting and packaging a collection of data to be received and unpackaged elsewhere in realtime.

OSC is an exceptionally powerful, lightweight, and flexible way of sending data between devices. It can be thought of as a more robust alternative to MIDI and supports a wide range of data types.

In Gesticulator, OSC is used to send motion sensor data from a device, as well as to receive and format the incoming data into Ableton Live.

Altogether, it works as follows:

1. ZIG SIM packages the live reading of your smartphone's motion sensors, then sends it all as a single bundle 30 times a second to Gesticulator.
2. Gesticulator receives, unbundles, and sends each sensor's data to its respective location in the Max for Live device.
3. The realtime readings of each sensors' axes by Gesticulator can then be used to modulate whichever parameter(s) within Ableton Live you choose to map to.

3.2 Sensors

Acceleration

The accelerometer sensor measures the force of acceleration on the device. Because gravity is always pulling on the device, a high-pass filter is automatically applied to the sensor to subtract the force of gravity (**g** = 9.81 m/s^2) and isolate the real acceleration of the device.[2]

This sensor is highly flexible when using the smoothing function, allowing both the most rapid and jumpy parameter changes of any sensor with smoothing off, as well as fluid sweeps when smoothing is increased.

Acceleration is best used when wanting to emphasize motion and is less ideal when precise or sustained values are required.

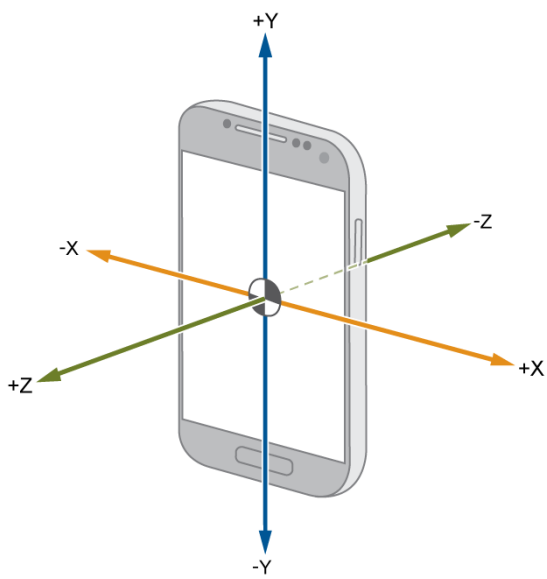

Fig. 4 The directions of the accelerometer's axes.[4]

Gravity

Gravity is a software sensor created via *sensor fusion* (combining two sensors together to produce a new sensor signal) using the gyroscope and accelerometer. This allows for the acceleration due to movement to be filtered out and leave only the pure gravity signal.

Typically, this sensor is used to determine the device's relative orientation in space.

This provides a much cleaner and more stable signal than the gyroscope alone and provides a three-dimensional vector indicating the direction each surface on your device is facing.

This is best used when precise or sustained values are desired, as well as fluid modulation between values.

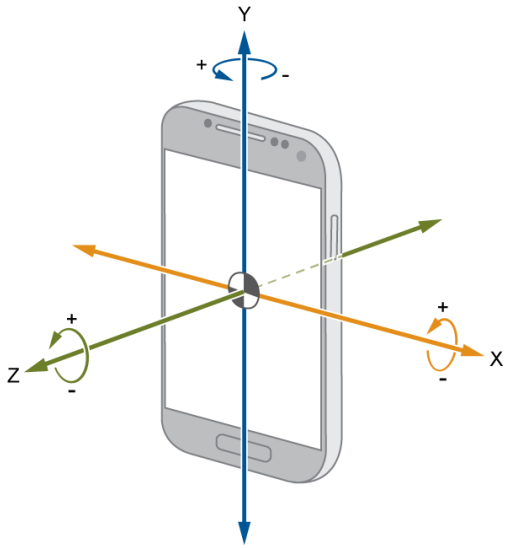

Fig. 5 The directions of the gravity sensor's axes.[4]

Quaternion

The quaternion represents the rotation of the device in 3D space. This is a more robust way to measure rotation than the gyroscope, as the quaternion can be used to rotate and scale the vector as movement occurs and thus produce a more accurate representation of the device's movement.

Unlike other sensors, this is represented in four axes: x, y, z, and w. The axis w is a product of the first three axes that allows for rotation and distance to be represented accurately.

This sensor is best used when wanting the closest representation of the device's motion in space by the mapped parameters. The four axes also provide the largest number of mapping possibilities of any sensor.

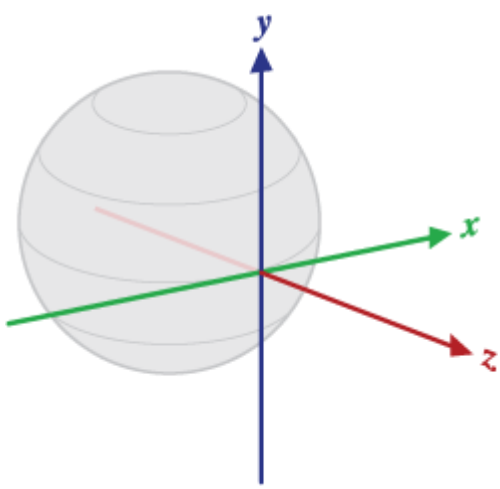

Fig. 6 The directions of the quaternion's real axes.[3]

Compass

The compass sensor measures the cardinal direction the device is pointing. Within Gesticulator, this is measured through the degrees of the compass (0 – 360°). Because there is only a single axis of measurement, there is only one mappable parameter for compass.

This sensor is best used when wanting to emphasize the direction you are facing. This is particularly useful for performances, where facing toward or away from the audience can produce radically different sonic results.

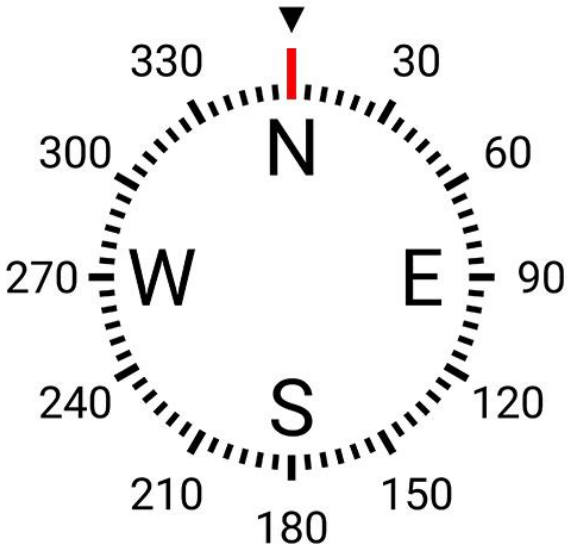

Fig. 7 The degrees of the compass corresponding to cardinal directions.

Touch

The touch sensor detects the position of 2D touch on the device's screen. The x-axis corresponds to the horizontal position of the touch, and the y-axis to the vertical position.

The visualizer for touch maps directly to the size and position of the screen, so that the bottom corresponds directly to the bottom of the screen, the top corresponds to the top of the screen, etc.

The touch sensor is great for getting precise values, dialing in parameters that are subsequently locked in regardless of motion, and modulating any two parameters as a matrix.

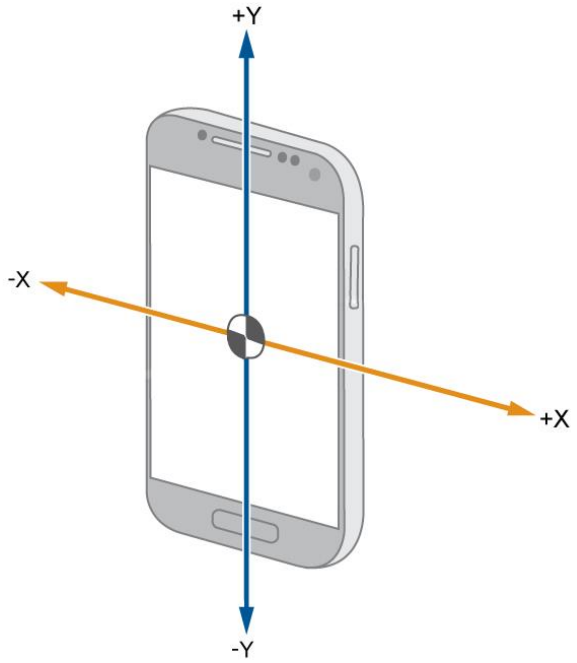

Fig. 8 The directions of the 2D touch sensor across the device's screen.

4. Troubleshooting

The device does not display input while ZIG SIM is sending:

1. Check that the device has correct network settings.
    a. Ensure the IP address being sent from the device is the same as the computer Gesticulator is running on.
        i. On Windows, open command prompt and type **ipconfig**.
        ii. Copy the address next to **IPv4 Address** to the sending device.
    b. Ensure the port being used is free and matches between the sending and receiving devices.
        i. On Windows, open command prompt and type **netstat -an**.
        ii. Look for the ports marked **UDP** and ensure the default port 50000 is listed. If it is, move to step 2.
        iii. If port 50000 is not listed, pick a different UDP port. Change the port in ZIG SIM listed under **Port Number** in the **Settings** tab, as well as in Gesticulator by clicking the **Port** button on the left of the device and entering the new port in the dialog.
2. Check firewall settings. Both "Max (32-bit)" and/or "Max (64-bit)" AND "Ableton Live 11 Suite" / "Ableton Live 10 Suite" need to be unblocked in your computer's firewall.
3. Use a new port in ZIG SIM and Gesticulator.
    a. Try other UDP ports listed in netstat.
    b. Try entering random 5-digit numbers (11111, 12345, etc.).
4. Change the device UUID in ZIG SIM Settings.
    a. Change the UUID to a new random number (2, 1234, etc.).

I'm using a device other than a smartphone, or an app other than ZIG SIM and the device does not display input for one or more parameters:

1. Check that the sensor data being sent is in the same format as ZIG SIM.
    a. Open the patcher window by clicking the Edit Button at the top of the device.
    b. Enter patching mode by clicking the unfreeze button ( ) and patching mode button ( ) in the bottom toolbar of the patch. Ensure the patch is unlocked ( ) in the bottom toolbar as well.
    c. Click the **red toggle** in the bottom left of the patch. This is provided to allow you to view the parameter names and formatting of any incoming OSC messages.
        i. If no messages are appearing in the connected message box, move on to step 2.
    d. Follow the instructions next to the toggle box to change the parameter names in the **route** object to fit the incoming OSC messages.

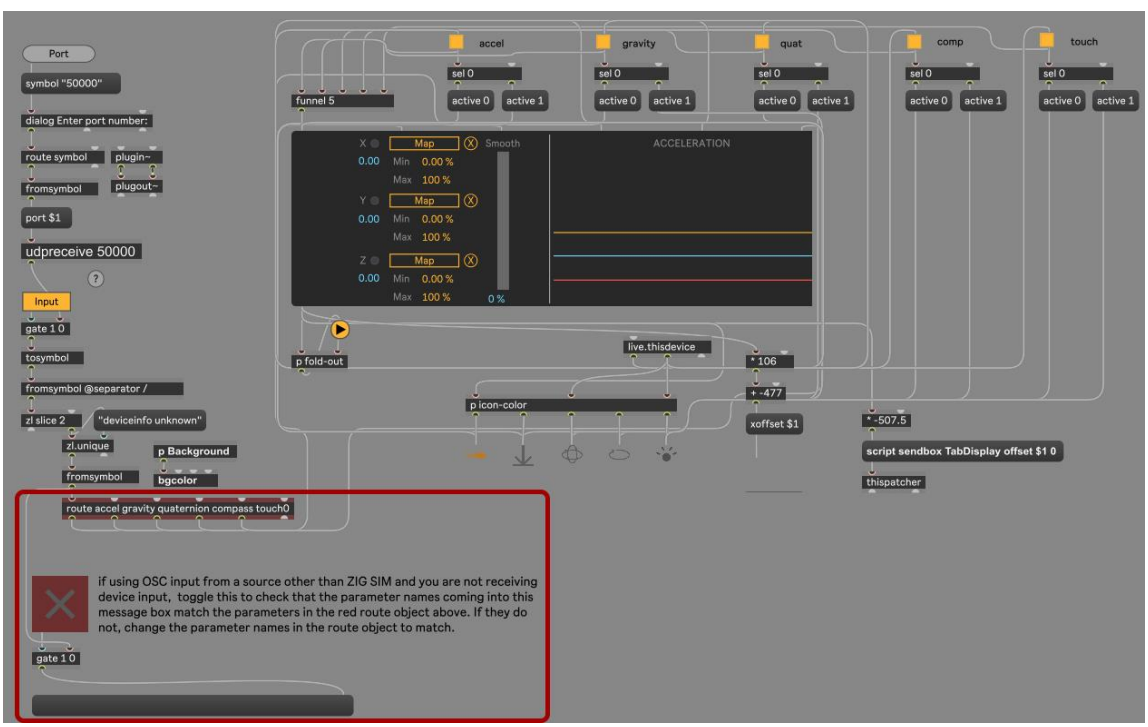

Fig. 9 The area for debugging the sensor input is indicated in the red square.

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    a. Ensure the IP address being sent from the device is the same as the computer Gesticulator is running on.
        i. On Windows, open command prompt and type **ipconfig**.
        ii. Copy the address next to **IPv4 Address** to the sending device.
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    a. Try other UDP ports listed in netstat.
    b. Try entering random 5-digit numbers (11111, 12345, etc.).
5. Change the device UUID in ZIG SIM Settings.
    a. Change the UUID to a new random number (1, 12345, etc.).

5. Credits

Gesticulator User Manual Version 1
for Windows and Mac

User Manual by Parish Bracha
Gesticulator created, designed, and programmed by Parish Bracha

Selected diagrams and reference material from:

[1] Stanford CCRMA OpenSoundControl

[2] Android Developers Manual

[3] Apple Developer Documentation

[4] Simulink Support Package for Android Devices

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